2.5 Classwork: Falling Objects Solutions

1. Dropped ball. Knowns: $v_0 = 0$, $y_0 = 0$, $a = +9.80$ m/s^2 (down positive).

   (a)

t (s)	0	1.00	2.00	3.00
v (m/s)	**0**	**9.80**	**19.6**	**29.4**
y (m)	**0**	**4.90**	**19.6**	**44.1**

   $v = at$ adds 9.80 m/s each second. $y = \frac{1}{2}at^2$: for example $\frac{1}{2}(9.80\ m/s^2)(2.00\ s)^2 = 4.90 \times 4.00 = 19.6\ m$. This is Giancoli's Example 2–10.

   (b) $\dfrac{19.6}{4.90} =$ **4 times as far**, and $\dfrac{44.1}{4.90} =$ **9 times as far**.
   The distance grows as the square of the time: twice the time gives $2^2 = 4$ times the distance, three times the time gives $3^2 = 9$ times.
   This is Galileo's result, $d \propto t^2$. Also accept "the ball falls farther each second": 4.90 m, then 14.7 m, then 24.5 m, in the ratio 1 : 3 : 5.

2. Leaning Tower of Pisa. Knowns: $v_0 = 0$, $y - y_0 = 55.86$ m, $a = +9.80$ m/s^2 (down positive).

   (a) Want t. Use $y = y_0 + v_0 t + \frac{1}{2}at^2$ with $v_0 = 0$:

   $$55.86\ \text{m} = \tfrac{1}{2}(9.80\ \text{m/s}^2)\,t^2, \quad \text{so } t^2 = \frac{2(55.86\ \text{m})}{9.80\ \text{m/s}^2} = 11.4\ \text{s}^2$$

   $t = \sqrt{11.4\ \text{s}^2} = 3.3763\ldots \approx$ **3.38 s**
   A common slip is to forget the $\frac{1}{2}$, giving $t^2 = 5.7$ and $t \approx 2.39$ s, or to stop at $t^2 = 11.4$.

   (b) Want v. Use $v = v_0 + at$, carrying the unrounded time:
   $v = 0 + (9.80\ \text{m/s}^2)(3.3763\ldots\ \text{s}) = 33.088\ldots \approx$ **33.1 m/s** downward
   Also accept $v^2 = v_0^2 + 2a(y - y_0) = 2(9.80)(55.86) = 1094.856$, so $v \approx 33.1$ m/s; this route does not need part (a). About 119 km/h.

   (c) **They land at the same time.** Without air resistance, every object falls with the same acceleration g, whatever its mass.
   Before Galileo, it was believed that a heavier object falls faster. For two dense balls the air makes almost no difference; it matters for a sheet of paper or a feather (Giancoli Figs. 2–20 and 2–21).

3. Ball thrown upward. Knowns: $v_0 = +14.7$ m/s, $a = -9.80$ m/s^2, $y_0 = 0$ at the hand (up positive).

   (a) **Velocity $\mathbf{v = 0}$** for an instant. **Acceleration $\mathbf{a = -9.80\ m/s^2}$**, the same as everywhere else in the flight.

   The common wrong answer is $a = 0$ at the top. Gravity does not switch off: if the acceleration were zero there, the velocity would stay zero and the ball would hang in the air (Giancoli Example 2–14).

   (b) Want t when $v = 0$. Use $v = v_0 + at$, so $t = \dfrac{v - v_0}{a}$:

   $t = \dfrac{0 - 14.7\text{ m/s}}{-9.80\text{ m/s}^2} = \mathbf{1.50\ s}$

   (c) Want $y - y_0$. Use $\bar{v} = \dfrac{v + v_0}{2} = \dfrac{0 + 14.7}{2} = 7.35$ m/s:

   $y - y_0 = \bar{v}\,t = 7.35\text{ m/s} \times 1.50\text{ s} = 11.025 \approx \mathbf{11.0\ m}$

   Also accept $v^2 = v_0^2 + 2a(y - y_0)$: $y - y_0 = \frac{0-(14.7)^2}{2(-9.80)} = \frac{-216.09}{-19.6} = 11.025$ m.

   (d) **3.00 s.** The fall back down takes as long as the rise: $2 \times 1.50\text{ s} = 3.00$ s.

   $v = v_0 + at = 14.7\text{ m/s} + (-9.80\text{ m/s}^2)(3.00\text{ s}) = \mathbf{-14.7\ m/s}$, that is, 14.7 m/s **downward**.

   The ball returns at the speed it was thrown, in the opposite direction; the minus sign carries the direction. Also accept setting $y = 0$ in $y = v_0 t + \frac{1}{2}at^2$: $0 = (14.7 - 4.90\,t)\,t$, so $t = 0$ or $t = 3.00$ s (Giancoli Example 2–13).

4. Raindrop. Knowns: $v_0 = 0$, $y - y_0 = 1000$ m, $a = +9.80$ m/s^2 (down positive).

   (a) Want v; no time is given. Use $v^2 = v_0^2 + 2a(y - y_0)$:

   $v^2 = 0 + 2(9.80\text{ m/s}^2)(1000\text{ m}) = 19{,}600\text{ m}^2/\text{s}^2$, so $v = \mathbf{140\ m/s}$

   About 500 km/h. Also accept finding $t = \sqrt{2(1000)/9.80} \approx 14.3$ s first, then $v = at \approx 140$ m/s.

   (b) **Air resistance.** The air pushes back on the drop, and the push grows as the drop goes faster, until it balances gravity. After that the drop stops speeding up and falls at a steady *terminal velocity*, about 9 m/s.

   Air resistance matters most for small, light objects with a large surface for their weight — like the flat sheet of paper in Giancoli Fig. 2–20. A raindrop reaches terminal velocity after falling only a few meters, so ignoring air here gives an answer more than 15 times too large.